LESSON 5.9

FINDING DISTANCES ON A COORDINATE PLANE

LESSON INTRODUCTION

MA.8.GR.1.2 Apply the Pythagorean theorem to solve mathematical and real-world problems involving the distance between two points in a coordinate plane.

LEARNING TARGETS

- I can determine the distance between two points on the coordinate plane.

BENCHMARK COHERENCE

BUILDING ON

MA.6.GR.1.2 Find distances between ordered pairs, limited to the same *x*-coordinate or the same *y*-coordinate, represented on the coordinate plane.

WORKING TOWARD

MA.912.GR.7.2 Given a mathematical or real-world context, derive and create the equation of a circle using key features.

ESSENTIAL QUESTIONS

- How can you find the distance of a diagonal line on the coordinate plane using the Pythagorean theorem?

LESSON NARRATIVE

In this lesson, students explore creating right triangles on the coordinate plane to assist them in finding the distance between two points on a diagonal line segment. The distance between two points will always be the hypotenuse of the student-drawn right triangle. Constructing the right triangle is similar to the process students will use to find slope between two points $\frac{\text{rise}}{\text{run}}$ in the next unit. The clarifications for the benchmark state that students should not memorize the distance formula, so it is not explicitly taught.

COMPONENT SUMMARY

5.9.1 WARM-UP *(~5 MIN.)*

Activate prior knowledge

5.9.2 EXPLORATION *(10 MIN.)*

Find distance between points on the coordinate plane

5.9.3 GUIDED PRACTICE *(15 MIN.)*

Find missing side lengths of triangles to determine if they are right triangles

5.9.4 GUIDED INSTRUCTION *(10 MIN.)*

Apply converse of Pythagorean theorem to determine if a triangle is a right triangle

5.9.5 YOUR TURN! *(5 MIN.)*

Practice applying distance on the coordinate plane and Pythagorean theorem

5.9.6 WRAP-UP *(~5 MIN.)*

Self-reflect on the learning target

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Hypotenuse
- Pythagorean theorem
- Converse of the Pythagorean theorem

MATERIALS

- (Optional) Ruler/straightedge
- (Optional) Graph paper
- Calculators

ADDITIONAL PREPARATION

- This is a vocabulary-rich lesson. Consider preparing your classroom-specific vocabulary or glossary strategies for support (word wall, math cards, etc.).

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may make errors when finding the vertical or horizontal distances on the coordinate plane.
- Students may have a rigid understanding of how to create the horizontal and vertical line segments, meaning they believe there is only one possible right triangle that can be created to determine the distance between two points.

THINGS TO CONSIDER

- Consider providing students with straightedges to help students draw their line segments.
- The lesson spirals students back through the converse of the Pythagorean theorem. Consider allowing students to use prior lessons to remind them of this theorem.
- If time is an issue, then consider assigning 5.9.5 Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

In 5.9.2 Exploration, question 5 lends itself to a Think-Pair-Share. Have students think independently first about how they would find the length of the diagonal. Prompt them to think about what we have learned in the last few lessons. Then students can discuss with their partner and share out with the class after deciding on a method.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

In this lesson, students are provided an opportunity to relate previously learned concepts to new concepts. Students relate what they know about the Pythagorean theorem to finding the missing hypotenuse of a triangle. Throughout the lesson, the scaffolding helps students create plans and procedures for finding the distance between two points by utilizing right triangles.

DIFFERENTIATE INSTRUCTION

Provide additional means of engagement and access through the use of digital manipulatives such as the Desmos online graphing calculator. This will be particularly helpful for students who struggle to graph points accurately.

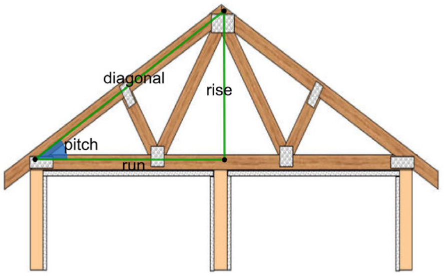
5.9.1 WARM-UP: NOTICE AND WONDER

TEACHER MOVES

Prompt students to make the connection between this question and the Pythagorean theorem, which we have been working on in the last few lessons.

5.9.1 Warm-Up: Notice and Wonder

In carpentry, right triangles are commonly used in constructing roofs. A roof's pitch is the steepness of the roof. It can be determined by the ratio between the two legs of the right triangle.

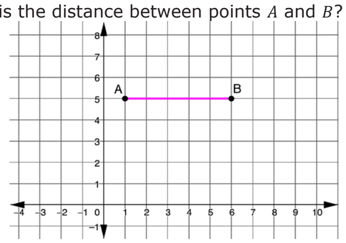


1. What do you notice or wonder about the use of right triangles in construction?
   Answers will vary. I notice that the triangles used in the picture are equilateral and are right triangles. I wonder what other types of triangles are commonly used in carpentry.

5.9.2 Exploration: Finding the Right Distance

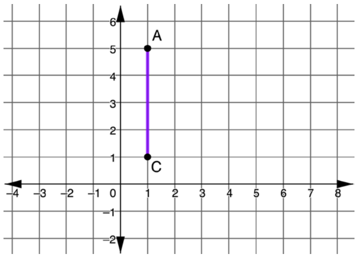
Work with your partner to answer each of the following.

1. What is the distance between points A and B?



5 units

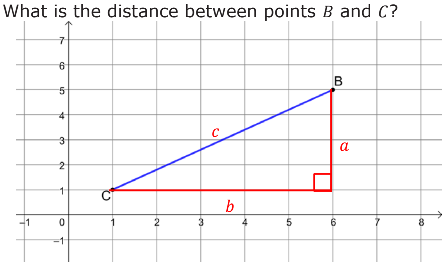
2. What is the distance between points A and C?



4 units

3. What strategy did you and your partner use to find the distance of the horizontal and vertical lines?
Answers will vary. We counted the boxes between the two points. We subtracted the coordinates of the two points.

4. What is the distance between points B and C?



   a. On the graph, label the length of segment BC, c.

   b. Create a right triangle with hypotenuse c by sketching horizontal and vertical legs. Label the legs a and b.

   c. Apply the Pythagorean Theorem to find the length of c. Round to the nearest tenth if necessary.
   6.4 units

5. Discuss with your partner how to find the length of any diagonal on the coordinate plane.

6. Complete the statement.

The length of any non-vertical or non-horizontal segment can be determined by using the endpoints as a [○ hypotenuse / ○ leg] of a right triangle and sketching the horizontal and vertical [○ hypotenuse / ○ legs] then applying the [○ Pythagorean Theorem. / ○ Area Formula.]

5.9.2 EXPLORATION: FINDING THE RIGHT DISTANCE

ENRICHMENT

Activate prior knowledge through questioning.

- How do you find the vertical distance between points on the coordinate plane?
- How do you find the horizontal distance between points on the coordinate plane?
- Which strategy makes the most sense to use?

TEACHER MOVES

Think-Pair-Share

Question 5 lends itself to a Think-Pair-Share.

Have students think independently first about how they would find the length of the diagonal. Prompt them to think about what we have learned in the last few lessons. Then students can discuss with their partner and share out with the class after deciding on a method.

5.9.3 GUIDED PRACTICE: FINDING DISTANCE ON THE COORDINATE PLANE

DIFFERENTIATE INSTRUCTION

Have students highlight and label the hypotenuse and legs on the triangle as they correspond to a, b, or c in the Pythagorean theorem. This will provide a visual entry point for students to access the material.

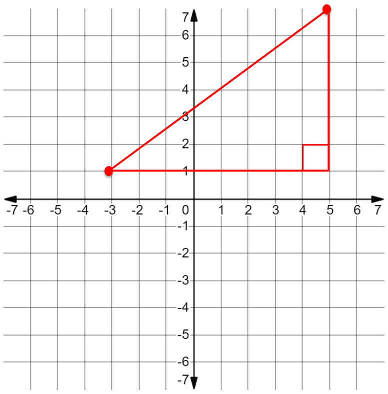
TEACHER MOVES

If students struggle to apply the Pythagorean theorem to find the missing side, ask prompting questions like:

- What information do we need to solve? (Horizontal and vertical distance of the legs and the hypotenuse.)
- How do you find distance when points are on different sides of the axis?

5.9.3 Guided Practice: Finding Distance on the Coordinate Plane

1. Use the coordinate grid to find the exact distance between $(-3,1)$ and $(5,7)$.



a. Plot and connect the points on the grid.

b. Create a right triangle by drawing horizontal and vertical line segments.

c. Apply the Pythagorean Theorem to find the missing side.

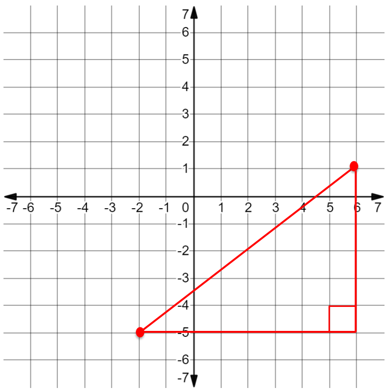
$$a^2 + b^2 = c^2$$

The exact distance between $(-3,1)$ and $(5,7)$ is 10 units.

TEACHER MOVES

Students may struggle to find the distance between the given points outside of the context of a right triangle. If you notice students struggling, encourage them to create a triangle on the grid to help visualize the problem. Students can then draw on the steps they have taken in previous problems to solve.

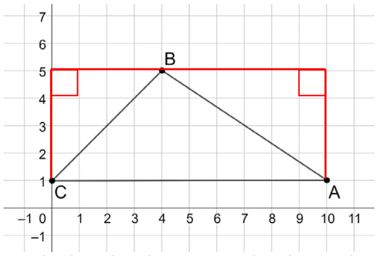
2. Use the coordinate grid to find the exact distance between $(-2,-5)$ and $(6,1)$.



The exact distance between $(-2,-5)$ and $(6,1)$ is 10 units.

5.9.4 Guided Instruction: Converse of the Pythagorean Theorem

The sketch shows a design for a new roof.



1. Predict whether this design is in the shape of a right triangle.
   Answers will vary. I predict that the design is in the shape of a right triangle. I predict that the design is not in the shape of a right triangle.

2. To determine whether this is a right triangle, determine the length of each side and prove whether the Pythagorean Theorem holds true.
   a. Find the length of $\overline{AC}$.
   $10\ units$
   b. Find the exact length of $\overline{AB}$ by creating a right triangle with $\overline{AB}$ as the hypotenuse.
   $\sqrt{52}\ units$
   c. Find the exact length of $\overline{BC}$ by creating a right triangle with $\overline{BC}$ as the hypotenuse.
   $\sqrt{32}\ units$
   d. List the three side lengths of the triangle.
   $\overline{AC}$ = ___10___ units
   $\overline{AB}$ = ___$\sqrt{52}$___ units
   $\overline{BC}$ = ___$\sqrt{32}$___ units

5.9.4 GUIDED INSTRUCTION: CONVERSE OF THE PYTHAGOREAN THEOREM

DIFFERENTIATE INSTRUCTION

Consider having either individual graphic organizers or an anchor chart with the steps for the Pythagorean theorem available throughout the lesson.

TEACHER MOVES

This section is intentionally scaffolded to guide students toward understanding how to use the converse of the Pythagorean theorem to determine if a triangle is a right triangle. Note that there is more than one right triangle that can be created with AB as the hypotenuse, but the measurements will be the same. This section is intentionally scaffolded to guide students towards understanding how to use the Converse of the Pythagorean Theorem to determine if a triangle is a right triangle. Note that there is more than one right triangle that can be created with AB as the hypotenuse, but the measurements will be the same.

DIFFERENTIATE INSTRUCTION

Consider highlighting vocabulary "converse of the Pythagorean theorem" along with a review of how the converse of the Pythagorean theorem works.

5.9.4 GUIDED INSTRUCTION: CONVERSE OF THE PYTHAGOREAN THEOREM (CONTINUED)

ENRICHMENT

If students struggle to explain whether Triangle ABC is a right triangle, further scaffold through questioning.

- How can we check if a triangle is a right triangle?
- What information do we need to know to apply the converse of the Pythagorean theorem?

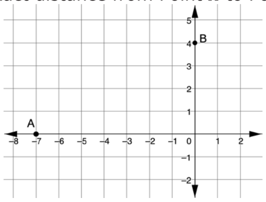
e. Determine if Triangle ABC is a right triangle using the converse of the Pythagorean Theorem.
Triangle ABC is not a right triangle because $10^2 = \sqrt{32}^2 + \sqrt{52}^2$ is not true.

3. Complete the statement.

The roof design ○ does / ○ does not satisfy the Pythagorean Theorem, therefore, the design ○ is / ○ is not a right triangle.

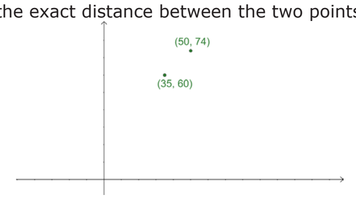
5.9.5 Your Turn!

1. Find the exact distance from Point A to Point B.



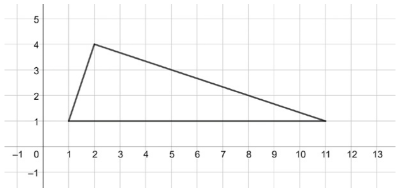
$\sqrt{65}$ *units*

2. Find the exact distance between the two points.



$\sqrt{421}$ *units*

3. Use the converse of the Pythagorean Theorem to determine if the figure is a right triangle.



The figure is a right triangle because $10^2 = \sqrt{10}^2 + \sqrt{90}^2$ *is true.*

5.9.5 YOUR TURN!

DIFFERENTIATE INSTRUCTION

Consider providing graph paper for students to use for question 2 or having students plot the points using the online Desmos graphing calculator. This will provide an additional layer of scaffolding for students who may still struggle to find the distance between diagonal points.

TEACHER MOVES

The right angle might not be immediately evident to students, and therefore students may struggle to determine which side is the hypotenuse. Provide further scaffolding through questioning. How can you use a corner of a sheet of paper to determine which angle is a right angle? Label it when you find it. Which side is the hypotenuse? How can you label the sides to accurately show the legs and hypotenuse? What is the converse of the Pythagorean theorem?

5.9.6 Wrap-Up: Reflection

1. Select the emoji face that corresponds to your feeling toward today's learning target.

I can determine the distance between two points on the coordinate plane.
Answers will vary.

5.9.6 WRAP-UP: REFLECTION

QUESTIONING

Students can go back and draw smiley faces on each activity to show how they felt throughout the individual activities. Challenge early finishers to explain the steps they take to find the distance between two points.